Finite Markov Decision Processes

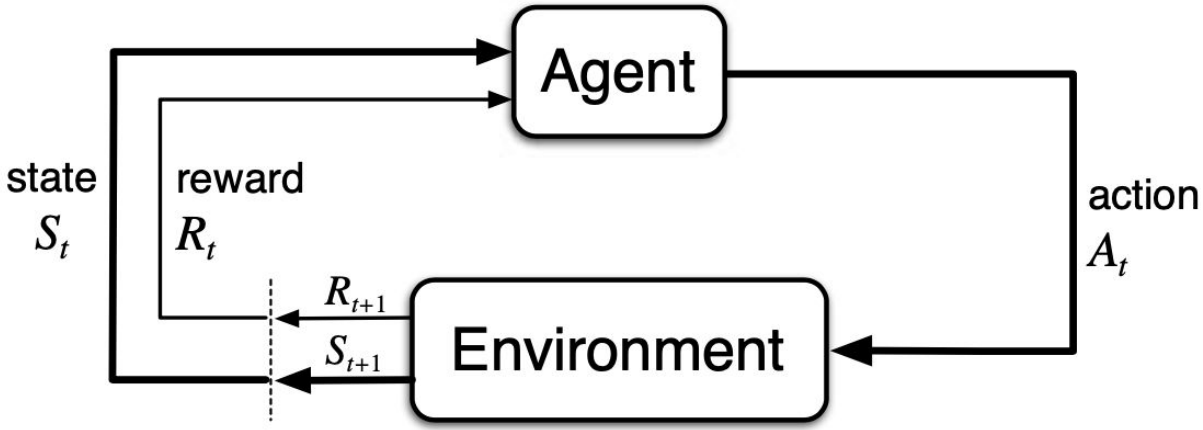


Figure 3.1: The agent–environment interaction in a Markov decision process.

Agent is the learner and the decision maker
Agent only performs actions
Environment rewards actions and indicates the new state

Agent and Environment

Actions influence not just immediate rewards but also *subsequent situations* and via them *future rewards*.

Some of what makes up a state could be based on memory of past sensations or even be entirely subjective (mental or imagined or computational).

- actions can be any decisions we want to learn how to make
- states are anything we can now that might be useful in making decisions

Agent — Environment Boundary

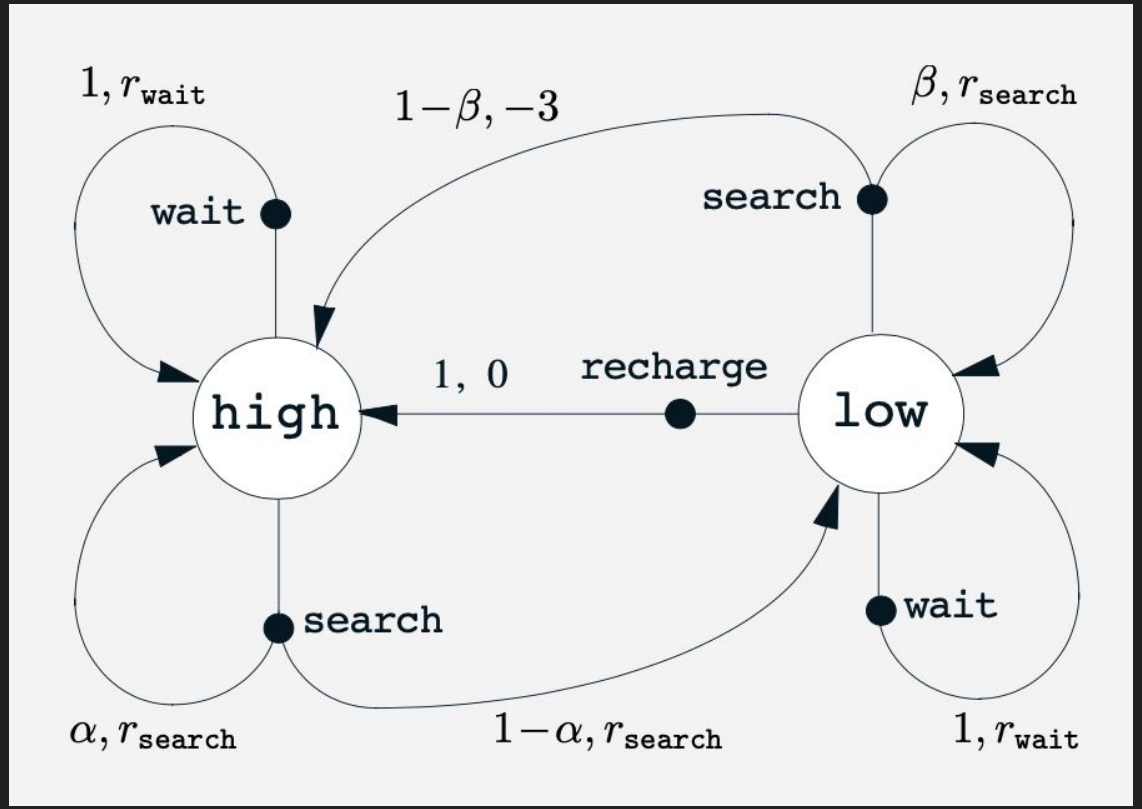
Agent-environment boundary represents the limit of the agent's *absolute control,* not of its *knowledge*

- Agent may know everything about the environment
- Yet it may find it difficult to learn the task
- Agent may be unable to solve the problem

Exercise 3.2

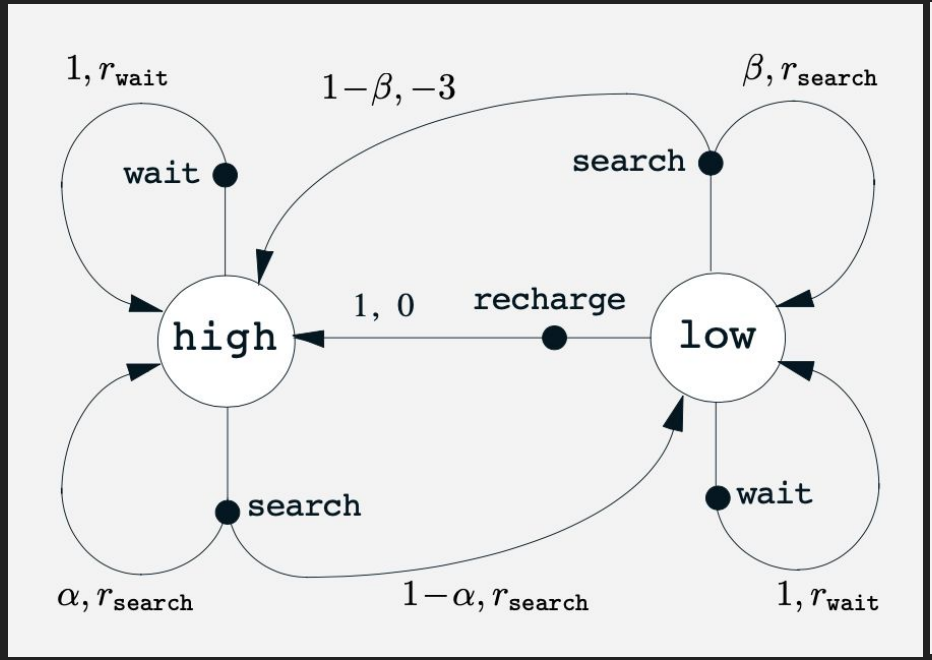
- Adversarial environments - noise.
- Pure MDPs cannot encode "must never happen" objectives.

Example 3.3 Recycling Robot



summarizing the dynamics of a finite MDP, as a *transition graph*. There are two kinds of nodes: *state nodes* and *action nodes*. There is a state node for each possible state (a large open circle labeled by the name of the state), and an action node for each state–action pair (a small solid circle labeled by the name of the action and connected by a line to the state node). Starting in state s and taking action a moves you along the line from state node s to action node (s, a). Then the environment responds with a transition to the next state's node via one of the arrows leaving action node (s, a). Each arrow corresponds to a triple (s, s', a), where s' is the next state, and we label the arrow with the transition probability, $p(s'|s, a)$, and the expected reward for that transition, $r(s, a, s')$. Note that the transition probabilities labeling the arrows leaving an action node always sum to 1.

Example 3.3 Recycling Robot



s	a	s'	$p(s' \mid s, a)$
high	search	high	α
high	search	low	$1-\alpha$
low	search	high	$1-\beta$
low	search	low	β
high	wait	high	1
high	wait	low	0
low	wait	high	0
low	wait	low	1
low	recharge	high	1
low	recharge	low	0

Finite MDP — *dynamics*

$$p(s', r \mid s, a) = \Pr\{ S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a \}$$

for all $s, s' \in S$, $r \in R$, and $a \in A(s)$.

$$t = 0, 1, 2, 3, \ldots$$

- At time step t
  - agent receives a representation of the environment state, S_t and on that basis
  - selects an action, A_t

Finite MDP — *dynamics*

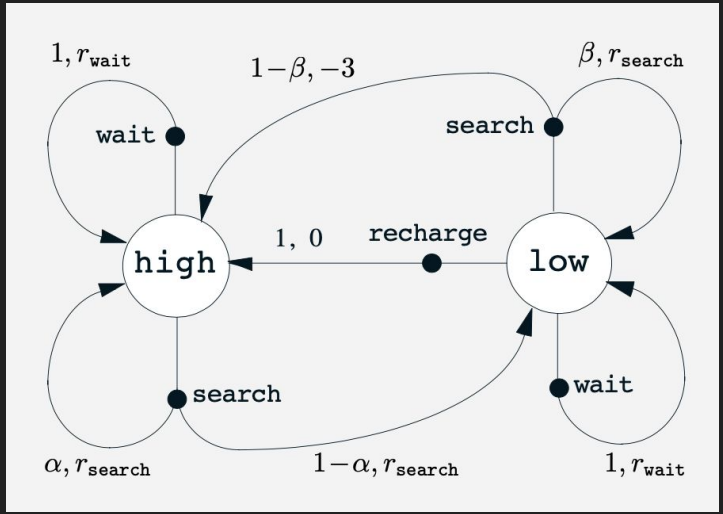
$$p(s', r \mid s, a) = \Pr\{ S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a \}$$

- The probability of each possible values for S_t and R_t depends on the immediately preceding state and action S_{t-1} and A_{t-1} and *not at all* on earlier states and actions.

- The state must include information about all aspects of the past agent-environment interaction that make a difference for the future.

Exercise 3.4

s	a	s'	r	$p(s', r \mid s, a)$
high	search	high	$r_{\texttt{search}}$	α
high	search	low	$r_{\texttt{search}}$	$1-\alpha$
low	search	high	-3	$1-\beta$
low	search	low	$r_{\texttt{search}}$	β
high	wait	high	$r_{\texttt{wait}}$	1
high	wait	low	$r_{\texttt{wait}}$	0
low	wait	high	$r_{\texttt{wait}}$	0
low	wait	low	$r_{\texttt{wait}}$	1
low	recharge	high	0	1
low	recharge	low	0	0



Finite MDP — *dynamics*

$$p(s', r \mid s, a) = \Pr\{ S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a \}$$

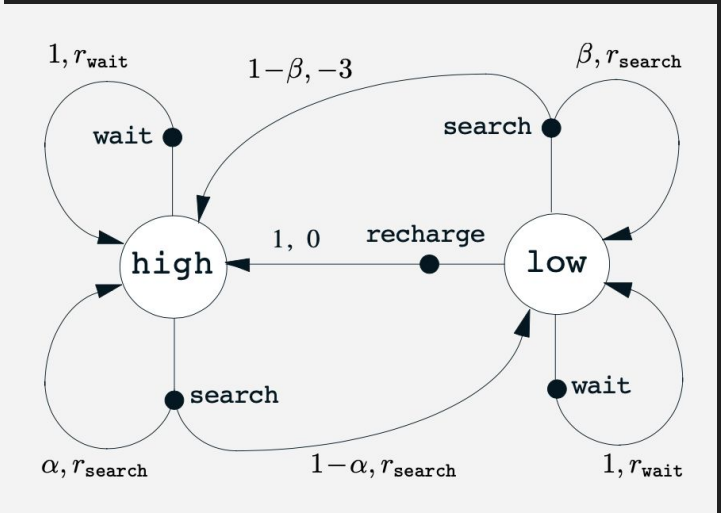
- Function p defines the *dynamics* of the MDP.

  p: S × R × S × A → [0, 1] is an ordinary deterministic function.

- p defines a probability distribution for each choice of s and a.

$$\sum_{s' \in S} \sum_{r \in R} p(s', r \mid s, a) = 1, \text{ for all } s \in S, a \in A(s)$$

s	a	s'	$p(s' \mid s, a)$	$r(s, a, s')$
high	search	high	α	$r_{\texttt{search}}$
high	search	low	$1-\alpha$	$r_{\texttt{search}}$
low	search	high	$1-\beta$	-3
low	search	low	β	$r_{\texttt{search}}$
high	wait	high	1	$r_{\texttt{wait}}$
high	wait	low	0	-
low	wait	high	0	-
low	wait	low	1	$r_{\texttt{wait}}$
low	recharge	high	1	0
low	recharge	low	0	-



Finite MDP — *state-transition probabilities*

$$p(s' \mid s, a) = \Pr\{ S_t = s' \mid S_{t-1} = s, A_{t-1} = a \}$$

$$= \sum_{r \in R} p(s', r \mid s, a)$$

Finite MDP — *state-transition probabilities*

$$p(s' \mid s, a) = \Pr\{ S_t = s' \mid S_{t-1} = s, A_{t-1} = a \}$$
$$= \sum_{r \in R} p(s', r \mid s, a)$$

s	a	s'	r	p
s_1	a_1	s_2	1	0.08
s_1	a_1	s_2	-10	0.10
s_1	a_1	s_2	15	0.31
s_1	a_1	s_2	-20	0.26

$$p(s_2 \mid s_1, a_1) = \sum_{r \in R} p(s_2, r \mid s_1, a_1)$$

Q: Should it always add up to 1?

Q: Why so?

Finite MDP — *state-transition probabilities*

$$p(s' \mid s, a) = \Pr\{ S_t = s' \mid S_{t-1} = s, A_{t-1} = a \}$$

$$= \sum_{r \in R} p(s', r \mid s, a)$$

- $p: S \times S \times A \rightarrow [0, 1]$ is a three argument function.

Finite MDP — *expected rewards for state-action pairs*

$$r(s, a) = \mathrm{E}[R_t \mid S_{t-1} = s, A_{t-1} = a]$$

$$= \sum_{r \in R} \mathrm{r} \sum_{s' \in S} p(s', r \mid s, a)$$

- $r: S \times A \rightarrow R$ is a two argument function.

Finite MDP — *expected rewards for state-action pairs*

$$r(s, a) = \mathrm{E}[R_t \mid S_{t-1} = s, A_{t-1} = a]$$

$$= \sum_{r \in R} r \sum_{s' \in S} p(s', r \mid s, a)$$

s	a	s'	r	p
s_1	a_1	s_2	2	0.08
s_1	a_1	s_3	-10	0.10
s_1	a_1	s_4	15	0.31
s_1	a_1	s_2	-20	0.26

$$2 \times \sum_{s' \in S} p(s', 2 \mid s_1, a_1) +$$

$$-10 \times \sum_{s' \in S} p(s', -10 \mid s_1, a_1) +$$

$$15 \times \sum_{s' \in S} p(s', 15 \mid s_1, a_1) +$$

$$-20 \times \sum_{s' \in S} p(s', -20 \mid s_1, a_1)$$

$$= ?$$

Finite MDP — *expected rewards for state-action pairs*

$$r(s, a) = \mathrm{E}[R_t \mid S_{t-1} = s, A_{t-1} = a]$$

$$= \sum_{r \in R} r \sum_{s' \in S} p(s', r \mid s, a)$$

s	a	s'	r	p
s_1	a_1	s_2	2	0.08
s_1	a_1	s_3	-10	0.10
s_1	a_1	s_4	15	0.31
s_1	a_1	s_2	-20	0.26

Q: Modify the transition graph by including two more states a few action-state transitions.

Finite MDP — *expected rewards for state-action pairs*

$$r(s, a) = \mathrm{E}[R_t \mid S_{t-1} = s, A_{t-1} = a]$$

$$= \sum_{r \in R} \mathrm{r} \sum_{s' \in S} p(s', r \mid s, a)$$

- $r: S \times A \rightarrow R$ is a two argument function.

MDP — *expected rewards for state-action-next_state triples*

$$r(s, a, s') = \mathrm{E}[R_t \mid S_{t-1} = s, A_{t-1} = a, s_t = s']$$

$$= \sum_{r \in R} r \frac{p(s', r \mid s, a)}{p(s' \mid s, a)}$$

s	a	s'	r	p
s_1	a_1	s_2	2	0.08
s_1	a_1	s_3	-10	0.10
s_1	a_1	s_4	15	0.31
s_1	a_1	s_2	-20	0.26

Q: *Compute* $r(s_1, a_1, s_2)$

Goals and Rewards

The return, denoted G_t, is defined as a specific function over the reward sequence.

$G_t = R_{t+1} + R_{t+2} + \ldots + R_T$ *where T is a final time step.*

- assumes a natural notion of final time-step, *or*
- agent-environment interaction is *episodic*
- Each episode ends in a terminal state followed by a reset to a
  - Standard starting state, or
  - Sample from the normal distribution of starting states

Different rewards for different outcomes in each episode.

$\boldsymbol{S}$ is the set of all nonterminal states

$\boldsymbol{S}^+$ is the set of all states plus the terminal state.